

EXACT TORSION GROUPS REALIZED ON GEOMETRICALLY SIMPLE GENUS-2 JACOBIANS OVER $\mathbb{Q}$

Groups are written by invariant factors: $[n_1, \dots, n_r]$ denotes $\mathbb{Z}/n_1\mathbb{Z} \times \dots \times \mathbb{Z}/n_r\mathbb{Z}$ with $n_1 \mid \dots \mid n_r$. Each row records one genus-2 curve $C/\mathbb{Q}$ whose Jacobian is geometrically simple and whose *full* rational torsion group is *exactly* the displayed group, $\text{Jac}(C)(\mathbb{Q})_{\text{tors}} \cong G$ (the first row is the trivial group, written $[\]$). The *Source(s)* column, by contrast, dates the *group*, and the earliest source there may establish only a rational point of the given order, a subgroup, or a positive-dimensional family containing G — not necessarily an exact-torsion or simple example; the companion document `torsion.sources.tex` classifies every source individually. We display, for each group, the equation of the curve of smallest conductor in the queried database snapshot (2026-07-22) with Jac geometrically simple and (where such a curve exists; see the endomorphism note) generic endomorphisms. Curves of small conductor lie in the production LMFDB; those found only in the extended database of Booker–Sutherland cite [5] in the *Source(s)* column. For convenience each equation is hyperlinked to its LMFDB home page: production curves by their permanent label, and curves that are only in the extended database via a alpha-site (<https://alpha.lmfdb.org>) equation-lookup URL, since alpha labels are not permanent (the lookup matches any isomorphic model of the curve). The equations are the intended stable identifiers.

Source(s) column. It gives the earliest construction of the torsion, together with the earliest certification of geometric simplicity when these differ; “new” means the exact simple realization first appears in the present paper. A dagger ($\dagger$) marks a source that exhibited the torsion but did not establish geometric simplicity of a realizing curve; in daggered rows the certification is supplied by the linked database entry or by the co-cited later source. Leprévost’s classical order-21, 25 and 27 curves were later shown non-simple or geometrically split, so those rows cite the first proven-simple realizations: the certified database curve 388.a.776.1 realizes [21], Nicholls [23] realizes [25], and for [27] both Nicholls [23] and the certified database curve 604.a.9664.2 (each distinct from Leprévost’s split curve [18]). Leprévost’s [3, 9] family is proven simple over $\mathbb{Q}$ only. Elkies’ page exhibits a rational order-31 *subgroup* with no rational generator (not a realization); the first *explicit* order-31 realization — previously the smallest cyclic order with no explicit realization, and the first on a genus-2 Jacobian — is the [31] row: the curve attached to the modular form 1830.2.a.q from the modular-abelian-surfaces dataset [7], whose exact rational torsion ($\mathbb{Z}/31\mathbb{Z}$, with generator of irrational Mumford support) and geometric simplicity are certified here. Order 31 also appears (for GL_2 -type abelian varieties, without curve equations) in the torsion conjectures of Alessandrì–Coppola [6], whose Theorem 4.2/4.3 proves an order-31 point exists on some inexplicit member of this isogeny class — the novelty here is the explicit curve, its exact group, and its certificates; their $g = 2$ lists include 37 as well, but at the one admissible level (2190, newform 2190.2.a.v) no member of the class is a Jacobian over $\mathbb{Q}$ — the real multiplication field $\mathbb{Q}(\sqrt{3})$ has narrow class number 2, so no member carries a rational principal polarization.

Endomorphism refinement. With two exceptions, every group here is realized by a curve with *generic* geometric endomorphisms, $\text{End}(\text{Jac}(C)_{\overline{\mathbb{Q}}}) = \mathbb{Z}$ (the displayed curve is the smallest-conductor known such witness). For database rows this is the LMFDB endomorphism certification (Costa–Mascot–Sijlsing–Voight). The exceptions, marked ^{RM}, are [2, 22] and [31]: no generic witness is known for either. For [2, 22] the extended database’s only geometrically simple curves have real multiplication ($\text{End}_{\overline{\mathbb{Q}}}$ a real quadratic order), so the displayed curve is geometrically simple but not generic. For [31] the only known witness is the displayed curve attached to the newform 1830.2.a.q (real multiplication by $\mathbb{Z}[\sqrt{2}]$, conductor $1830^2 = 3348900$, outside the extended database’s conductor range); its 31-torsion arises from an Eisenstein congruence at the level prime 61 ($31 \mid 61 + 1$). For [2, 2, 14] the extended database likewise contains only RM witnesses; the displayed curve — of conductor $2^2 3^2 5^2 7^2 13^2 \cdot 19 \cdot 23 \cdot 37 \cdot 317 = 38200383011700$, the smallest of ten generic witnesses found here by a three-root refinement of a contact-7 construction

— is new to this paper. For each displayed non-database equation with generic endomorphisms (the rows marked “new” without an ^{RM} flag, and the $[2, 2, 2, 10]$ family member) we give a self-contained certificate in `data/claude_endz_certificates.txt`: a good prime whose L -polynomial is irreducible with all Frobenius-root powers of degree 4 (geometric simplicity, by the root-power criterion), a second good prime whose irreducible L -polynomial has splitting field linearly disjoint from the first (so the centre of the geometric endomorphism algebra is $\mathbb{Q}$, excluding real and complex multiplication; cf. [27, 28]), and the observation that $\# \text{Jac}(C)(\mathbb{Q})_{\text{tors}} > 18$, which excludes quaternionic multiplication since an (O-)PQM genus-2 Jacobian over $\mathbb{Q}$ has at most 16 (resp. 18) rational torsion points [29]. The RM-flagged non-database row [31] cannot carry such an $\text{End} = \mathbb{Z}$ certificate (its geometric endomorphism ring is $\mathbb{Z}[\sqrt{2}]$); its dedicated certificate — exact torsion invariants, the D_4 /root-power geometric-simplicity witness at $p = 11$, and the real-multiplication identification (constant Frobenius discriminant core 2 at all 164 good primes $p \leq 1000$, Hecke-trace match to 1830.2.a.q) — is in `data/claude_z31_rm_certificate.txt`. The one caveat is that the earliest cited simple realization is occasionally itself non-generic — ‡ marks [5], whose first simple realizations are the RM Jacobian $J_0(31)$ and the CM curve $y^2 - y = x^5$; the displayed witness in every such row is nonetheless generic.

Group	Curve	Source(s)
[]	$y^2 + y = -18x^6 - 637x^5 - 3823x^4 + 31054x^3 - 63684x^2 + 51520x - 14580$	[4]
[2]	$y^2 + (x^2 + x + 1)y = x^5 - 40x^3 + 22x^2 + 389x - 608$	[4]
[3]	$y^2 + y = x^5 - 2x^4 + 2x^3 - x^2$	[4]
[4]	$y^2 + (x^3 + 1)y = x^3 - x$	[4]
[5]	$y^2 + y = x^5 - 9x^4 + 14x^3 - 19x^2 + 11x - 6$	[21] [†] , [3] [‡]
[6]	$y^2 + (x^2 + x)y = x^5 - 12x^4 + 26x^3 + 46x^2 + 21x + 3$	[11] [†]
[7]	$y^2 + x^3y = x^5 - 3x^3 + 3x - 2$	[21] [†] , [20]
[8]	$y^2 + (x + 1)y = -x^6 - 2x^5 - 2x^4 - x^3$	[4]
[9]	$y^2 + (x^3 + x + 1)y = -x^4$	[11] [†]
[10]	$y^2 + (x^3 + x)y = x^5 - 2x^4 - 8x^3 + 16x + 7$	[11] [†]
[11]	$y^2 + (x^3 + x + 1)y = x^2$	[21] [†] , [2]
[12]	$y^2 + (x^3 + x^2 + x)y = x^2 + x + 1$	[4]
[13]	$y^2 + (x^3 + x^2 + x + 1)y = -x^3 - x^2$	[10, 15] [†] , [26]
[14]	$y^2 + (x^3 + 1)y = x^2 + x$	[25, 26]
[15]	$y^2 + (x^3 + x^2 + x + 1)y = -x^2 - x$	[16] [†]
[16]	$y^2 + (x^3 + 1)y = -x^5 + x^4 - 2x^2 + x + 1$	[4]
[17]	$y^2 + (x^3 + x^2 + x)y = x^3 + x^2 + 3x + 1$	[16] [†] , [26]
[18]	$y^2 + (x^3 + 1)y = -2x^4 + 4x^2 + 2x$	[25, 26]
[19]	$y^2 + (x^2 + x + 1)y = x^6 + x^5 + x^4 - 5x^3 - 3x^2 + 5x - 2$	[16] [†] , [26]
[20]	$y^2 + (x + 1)y = -x^5$	[19] [†]
[21]	$y^2 + (x^3 + x + 1)y = -x^4 + 2x^2 + x$	[16] [†] , [23]
[22]	$y^2 + (x^3 + x)y = x^3 - 2x^2 - x + 1$	[17] [†] , [18]
[23]	$y^2 + x^2y = x^6 - x^5 - 2x^4 + x^3 + 2x^2 - 2x + 1$	[24] [†] , [18]
[24]	$y^2 + (x^3 + 1)y = 2x^4 + 3x^3 + 4x^2 + 2x$	[17] [†] , [18]
[25]	$y^2 + (x + 1)y = -9x^5 + 25x^4 + 75x^3 + 41x^2 - 23x + 2$	[17] [†] , [23]
[26]	$y^2 + (x + 1)y = -12x^5 - 3x^4 - 3x^3 - 3x^2$	[17] [†] , [18]
[27]	$y^2 + (x^3 + 1)y = -x^4 + x^3 + x^2 - x$	[14] [†] , [23]
[28]	$y^2 + (x^3 + 1)y = -x^5 + x^3 + x^2 + 3x + 2$	[25, 26]
[29]	$y^2 + (x + 1)y = x^6 - 2x^5 + 2x^3 - x^2$	[17] [†] , [18]

Group	Curve	Source(s)
[30]	$y^2 + (x^3 + x^2)y = 2x^6 + 4x^5 - 10x^4 - 12x^3 + 16x^2 - x$	[22]
[31]	$y^2 + (-x^2 - x)y = -839x^6 + 2841x^5 - 4587x^4 + 4300x^3 - 2466x^2 + 816x - 126$	[7] [†] , new ^{RM}
[32]	$y^2 + (x^2 + x)y = x^6 - x^5 + x^4 + 45x^3 + 49x^2 - 49x - 47$	[8]
[33]	$y^2 + (x^2 + x + 1)y = x^6 - 7x^5 + 9x^4 + 7x^3 + 21x^2 + 13x + 2$	[26]
[34]	$y^2 + (x^3 + x^2)y = -2x^5 + 3x^3 + 7x^2 + 12x + 15$	[8]
[36]	$y^2 + x^2y = 3x^5 + 30x^4 + 13x^3 - 14x^2 + 6x + 9$	[22]
[39]	$y^2 + (x^3 + 1)y = x^4 + 2x^3 + x^2 - x$	[8]
[40]	$y^2 + (x + 1)y = 3x^5 + 4x^4 - 2x^3 - 2x^2$	[8]
[2, 2]	$y^2 + xy = 4x^5 + 33x^4 + 72x^3 + 16x^2 + x$	[4]
[2, 4]	$y^2 + xy = x^5 - 8x^4 + 16x^3 - x$	[4]
[2, 6]	$y^2 + y = 4x^5 + 4x^4 - x^3 - 2x^2$	[4]
[2, 8]	$y^2 + (x + 1)y = 8x^5 + 3x^4 - 4x^3 - 2x^2$	[4]
[2, 10]	$y^2 + (x + 1)y = 3x^5 - 2x^4 - 4x^3 + x^2 + x$	[4]
[2, 12]	$y^2 + (x^2 + x)y = x^5 - 8x^4 + 14x^3 + 2x^2 - x$	[4]
[2, 14]	$y^2 + (x^3 + x)y = -2x^4 - x^3 + x + 1$	[4]
[2, 16]	$y^2 + (x^3 + x^2)y = -x^5 - x^4 + x^3 - 3x^2 + 4x + 8$	[5]
[2, 18]	$y^2 + (x^2 + x)y = x^6 - x^5 - 5x^4 + 3x^3 + x^2 + 2x + 7$	[5]
[2, 20]	$y^2 + x^2y = 4x^6 - 30x^5 + 28x^4 + 185x^3 - 251x^2 - 180x$	[5]
[2, 22]	$y^2 + (x^2 + x)y = x^6 - 3x^5 + 9x^4 - 5x^3 + 12x^2 - 6x$	[5] ^{RM}
[2, 26]	$y^2 + (x^2 + 1)y = 9x^6 - 3x^5 - 12x^4 + 3x^3 + 2x^2$	[5]
[2, 28]	$y^2 + (x^2 + 1)y = 9x^6 - 3x^5 - 42x^4 + 34x^3 + 4x^2 - 3x$	[5]
[3, 3]	$y^2 + y = x^6 + 4x^4 + 2x^3 + 7x^2 + 3x + 3$	[1]
[3, 6]	$y^2 + (x^3 + x^2 + x + 1)y = -x^5 - 5x^3 + 7x^2 - 2x + 21$	[5]
[3, 9]	$y^2 + (x^3 + 1)y = 16x^6 + 75x^5 + 40x^4 - 113x^3 - 230x^2 - 75x + 326$	[18] [†] , [5]
[4, 4]	$y^2 + (x^2 + x)y = 99x^6 + 242x^5 + 1823x^4 + 2601x^3 + 10832x^2 + 6534x + 20493$	[5]
[4, 8]	$y^2 + (x^2 + x)y = -81x^5 + 272x^4 - 435x^3 + 450x^2 + 682x + 164$	[5]
[6, 6]	$y^2 = x(39x^2 - 69x + 125)(48x^2 + 8x + 39)$	new
[2, 2, 2]	$y^2 + xy = 10x^5 + 8x^4 - 5x^3 - 3x^2 + x$	[4]
[2, 2, 4]	$y^2 + (x^2 + x)y = x^5 + 3x^4 - 3x^3 - 8x^2 + 6x$	[4]
[2, 2, 6]	$y^2 + (x^2 + 1)y = 3x^5 - 4x^3 - x^2 + x$	[4]
[2, 2, 8]	$y^2 + (x^2 + x)y = 9x^5 + 20x^4 - 3x^3 - 6x^2 - x$	[5]
[2, 2, 10]	$y^2 + (x^2 + 1)y = -12x^5 + 6x^4 + 12x^3 - 7x^2$	[5]
[2, 2, 12]	$y^2 + (x^2 + x)y = x^6 - 7x^5 + 8x^4 + 16x^3 - 5x^2 - 5x$	[5]
[2, 2, 14]	$y^2 = (x + 1)(x + 9)(2x - 115)(6x + 55)(3x^2 - 220x + 2652)$	[5], new
[2, 2, 20]	$y^2 = -(x - 1)(6x + 1)(2x + 7)(6217x + 1008)(21x^2 - 161x + 144)$	new
[2, 4, 4]	$y^2 + (x^2 + x)y = -116x^5 + 2204x^4 - 1850x^3 - 210x^2 + 180x$	[5]
[2, 4, 8]	$y^2 = x(3x - 5)(5x + 333)(16x^2 + 23x + 576)$	new
[2, 2, 2, 2]	$y^2 + (x^2 + x)y = -27x^5 + 23x^4 + 8x^3 - 4x^2 - x$	[5]
[2, 2, 2, 4]	$y^2 + (x^2 + x)y = -225x^5 + 146x^4 - 16x^3 - 6x^2 + x$	[5]
[2, 2, 2, 6]	$y^2 + (x^2 + 1)y = -6x^5 - 14x^4 + 18x^3 + 11x^2 - 12x + 2$	[5]
[2, 2, 2, 8]	$y^2 = x(x + 1)(x + 55^2)(x + 99^2)(x + 125^2)$	new
[2, 2, 2, 10]	$y^2 = x(x + 1)(x - 1)(3x - 7)(8x - 13)(24x + 25)$	[9]
[2, 2, 2, 12]	$y^2 = x(x - 120^2)(x - 143^2)(x - 266^2)(x - 218^2)(x - 241^2)$	new

Group	Curve	Source(s)
[2, 2, 4, 4]	$y^2 = x(x + 36^2)(x + 57^2)(x + 64^2)(x + 132^2)$	new

REFERENCES

- [1] N. Bruin, E. V. Flynn, and D. Testa, *Descent via (3,3)-isogeny on Jacobians of genus 2 curves*, Acta Arith. **165** (2014), 201–223.
- [2] N. Bernard, F. Leprévost, and M. Pohst, *Jacobians of genus-2 curves with a rational point of order 11*, Experiment. Math. **18** (2009), 65–70.
- [3] J. Boxall and D. Grant, *Examples of torsion points on genus two curves*, Trans. Amer. Math. Soc. **352** (2000), 4533–4555.
- [4] A. R. Booker, J. Sijsling, A. V. Sutherland, J. Voight, and D. Yasaki, *A database of genus-2 curves over the rational numbers*, LMS J. Comput. Math. **19(A)** (2016), 235–254.
- [5] A. R. Booker and A. V. Sutherland, *Genus 2 curves of small conductor*, in preparation.
- [6] J. Alessandrì and N. Coppola, *Torsion points on GL_2 -type abelian varieties*, arXiv:2602.21047 (v3, April 2026); data repository <https://github.com/NirvanaC93/Torsion-of-GL2-abelian-varieties>.
- [7] E. Costa, N. D. Elkies, S. Hashimoto, R. Jha, K. Martin, B. Poonen, and J. Voight, *Modular abelian surfaces* (data repository), <https://github.com/edgarcosta/ModularAbelianSurfaces>, 2024.
- [8] N. D. Elkies, *Curves of genus 2 over $\mathbb{Q}$ whose Jacobians are absolutely simple abelian surfaces with torsion points of high order*, https://people.math.harvard.edu/~elkies/g2_tors.html, 2001–2002, updated 2010.
- [9] N. D. Elkies, *Families of genus-2 curves with 5-torsion*, LuCaNT: LMFDB, Computation, and Number Theory, Contemp. Math. **796**, Amer. Math. Soc., 2024, 165–185.
- [10] E. V. Flynn, *Large rational torsion on abelian varieties*, J. Number Theory **36** (1990), 257–265.
- [11] E. V. Flynn, *Sequences of rational torsions on abelian varieties*, Invent. Math. **106** (1991), 433–442.
- [12] E. W. Howe, F. Leprévost, and B. Poonen, *Sous-groupes de torsion d'ordres élevés de Jacobiennes décomposables de courbes de genre 2*, C. R. Acad. Sci. Paris Sér. I **323** (1996), 1031–1034.
- [13] E. W. Howe, F. Leprévost, and B. Poonen, *Large torsion subgroups of split Jacobians of curves of genus two or three*, Forum Math. **12** (2000), 315–364.
- [14] E. W. Howe, *Genus-2 Jacobians with torsion points of large order*, Bull. London Math. Soc. **47** (2015), 127–135.
- [15] F. Leprévost, *Famille de courbes de genre 2 munies d'une classe de diviseurs rationnels d'ordre 13*, C. R. Acad. Sci. Paris Sér. I Math. **313** (1991), 451–454.
- [16] F. Leprévost, *Familles de courbes de genre 2 munies d'une classe de diviseurs rationnels d'ordre 15, 17, 19 ou 21*, C. R. Acad. Sci. Paris Sér. I Math. **313** (1991), 771–774.
- [17] F. Leprévost, *Points rationnels de torsion de jacobiniennes de certaines courbes de genre 2*, C. R. Acad. Sci. Paris Sér. I **316** (1993), 819–821.
- [18] F. Leprévost, *Jacobiniennes de certaines courbes de genre 2: torsion et simplicité*, J. Théor. Nombres Bordeaux **7** (1995), 283–306.
- [19] F. Leprévost, *Sur certains sous-groupes de torsion de jacobiniennes de courbes hyperelliptiques de genre $g \geq 1$* , Manuscripta Math. **92** (1997), 47–63.
- [20] F. Leprévost, M. Pohst, and A. Schöpp, *Rational torsion of $J_0(N)$ for hyperelliptic modular curves and families of Jacobians of genus 2 and genus 3 curves with a rational point of order 5, 7 or 10*, Abh. Math. Sem. Univ. Hamburg **74** (2004), 193–203.
- [21] A. Ogg, *Rational points on certain elliptic modular curves*, Proc. Sympos. Pure Math. XXIV, Amer. Math. Soc., 1973, 221–231.
- [22] V. P. Platonov and M. M. Petrunin, *New curves of genus 2 over the field of rational numbers whose Jacobians contain torsion points of high order*, Dokl. Math. **91** (2015), 220–221.
- [23] C. Nicholls, *Descent methods and torsion on Jacobians of higher genus curves*, Ph.D. thesis, University of Oxford, 2018.
- [24] H. Ogawa, *Curves of genus 2 with a rational torsion divisor of order 23*, Proc. Japan Acad. Ser. A **70** (1994), 295–298.
- [25] V. P. Platonov and M. M. Petrunin, *New orders of torsion points in Jacobians of curves of genus 2 over the rational number field*, Dokl. Math. **85** (2012), 286–288.
- [26] V. P. Platonov, V. S. Zhgun, and M. M. Petrunin, *On the simplicity of Jacobians for hyperelliptic curves of genus 2 over the field of rational numbers with torsion points of high order*, Dokl. Math. **87** (2013), 318–321.
- [27] E. Costa, D. Lombardo, and J. Voight, *Identifying central endomorphisms of an abelian variety via Frobenius endomorphisms*, Res. Number Theory **7** (2021), no. 46.
- [28] D. Zywina, *Determining monodromy groups of abelian varieties*, Res. Number Theory **8** (2022), no. 89.
- [29] J. Laga, C. Schembri, A. Shnidman, and J. Voight, *Rational torsion points on abelian surfaces with quaternionic multiplication*, Forum Math. Sigma **12** (2024), e92.